University of Bahrain
College of Information Technology

Senior Project Proposal

UNDER PROCESS

Project title: *19. Myhealth : Develop an application to manage health records, appointments, and medical staff with the power of AI*

Student 1: Ali Mohamed Jaafar Mohamed,202208244, 202208244@stu.uob.edu.bh
Student 2: Mohammed A.Redha Meftah, 202209027, 202209027@stu.uob.edu.bh
Student 3: , ,

Supervisor Name: Dr. Amal Ghanim
Co-supervisor Name:

Academic Year: 2025/2026, **Semester:** Second

Submitted on: 1/2/2026 7:48:53 PM

---

1. Problem Statement

Healthcare systems today are slow and complicated, causing problems for everyone. Patients can't find their health records easily because they are scattered across different hospitals and clinics. Booking an appointment is a frustrating, manual process that leads to long wait times and missed visits. At the same time, doctors and nurses are buried in paperwork instead of focusing on patient care. MyHealth AI is an application designed to solve these problems by using artificial intelligence. The app gathers all of a patient's medical records into one secure, easy-to-read timeline. It also automates appointment booking, suggesting the best available times and sending smart reminders to reduce no-shows. For medical staff, the app provides a clear dashboard that uses AI to highlight urgent cases, prioritize daily tasks, and spot health trends—helping them make faster, better decisions. By bringing everything together and adding intelligent automation, this system will reduce wait times, cut down on administrative work, improve patient care, and make healthcare more efficient for everyone.

2. Project Objectives

Research Question 1: How can Artificial Intelligence synthesize fragmented health records into a coherent, actionable patient summary? Objective: Develop an AI module to ingest heterogeneous data (PDF reports, clinician notes) and generate a unified, timeline-based health profile with highlighted key events and trends. Research Question 2: How can an AI-driven scheduler optimize booking to reduce no-shows and maximize clinic resource efficiency? Objective: Implement an intelligent scheduling system that uses predictive models (based on history, type of visit) to recommend optimal slots and manage dynamic reminders. Research Question 3: How can AI assist medical staff in daily task prioritization and early identification of patient risks?

3. Project Significance

This project can make healthcare much better for everyone involved. For patients, it means having all your health information in one easy-to-use place and booking appointments without hassle. For hospitals and clinics, it helps things run smoother, saves money, and makes better use of time and resources. For doctors and nurses, the AI tools can reduce stress, help prevent errors, and allow them to focus more on preventing health issues. By putting all of this into one system, MyHealth AI is a practical solution that can improve health results, make patients happier, and lower overall costs, moving healthcare toward a more modern and efficient future.

4. Brief Literature Review

Research shows that current healthcare systems struggle with disconnected information and inefficient processes, which is exactly what our project aims to solve. Recent studies confirm that AI can effectively summarize scattered patient records into clear timelines (Lee et al., 2023) and that smart scheduling systems using patient data can significantly reduce missed appointments (Miller & Park, 2022). Furthermore, evidence indicates that AI tools help medical staff manage their workload and spot urgent cases faster (Garcia et al., 2024). While useful tools exist for individual tasks—like patient portals or booking apps—there is a lack of a single platform that uses AI to link all these functions together for both patients and staff. Our project, MyHealth AI, is designed to fill this gap by integrating these proven AI capabilities into one system.

5. Supportive References

1. Lee, S., et al. (2023). Automated Summarization of Clinical Notes via Natural Language Processing: Improving Provider Efficiency. Journal of Biomedical Informatics. 2. Miller, A., & Park, S. (2022). A Machine Learning Framework for Predicting Outpatient No-Shows to Optimize Scheduling. AMIA Annual Symposium Proceedings. 3. Garcia, M., et al. (2024). Evaluating an AI Clinical Decision Support Tool for Nurse Workflow Prioritization in Hospital Settings. International Journal of Medical Informatics.

6. Project Timeline (starting weeks)

Requirement Collection: W 1-3
Literature Review: W 1-3
Design: W 4-6
Implementation: W 7-9
Testing and Results: W 10-12
Report Writing: W 13-16
Presentation: W 13-16

---

7. Statement of Authenticity

Plagiarism is the act of copying, imitating, or asking for major help from others to develop one or more parts of the senior project. Any attempt of plagiarism will be reported and the student(s) will be penalized according to the University of Bahrain rules.
Upon the submission of this document, we, the students of this project, declare that we will submit an original work of our own for all the parts of the project and we will explicitly acknowledge any work of others. In addition to that, we will ensure that the project to be submitted had not been submitted for any other course.

---

8. Committee's Decision

{not finalized}

---

9. Head of the department's Decision

{If required}.